

A Topological Model-Driven Optimization Approach for Detecting and Locating Electricity Fraud in low-voltage Networks

Rubén Carmona-Pardo^{*†}, Rafael Morán-Corbacho^{*†}, Carlos García-Santacruz^{*†}, Álvaro Rodríguez del Nozal^{*}, and Esther Romero-Ramos^{*†}

^{*} Department of Electrical Engineering, Universidad de Sevilla, Seville, Spain

[†] Ingelectus Innovative Electrical Solutions, Seville, Spain

[‡] ENGREEN Laboratory of Engineering for Energy and Environmental Sustainability of Universidad de Sevilla

Abstract—Electricity fraud in low-voltage networks has become increasingly sophisticated, causing both technical and non-technical losses for utilities, along with issues related to power quality and supply continuity. While current artificial intelligence and big data methods are effective for detecting visible fraud, invisible fraud (resulting from direct illegal connections) is harder to detect and locate, often requiring extensive fieldwork. This paper proposes a methodology that leverages network topology, connectivity data, and smart meter measurements to construct a linear model of the low-voltage network and address both visible and invisible fraud. The optimization problem minimizes the differences between recorded and calculated voltages, as well as the discrepancies in power, to detect and locate fraudulent segments. This approach significantly reduces the time required for fraud detection, optimizing efforts and reducing costs for distribution operators. Real-world low-voltage network data tests validate the proposal.

Index Terms—Electricity fraud, Fraud detection, low-voltage networks, Network optimization, Smart Meters

NOMENCLATURE

Throughout this work, the following notation is used:

k, t	Indices for buses
$p, q \in \{a, b, c\}$	Indices for phases
V_k^q	Phase q to neutral voltage magnitude measurement at bus k
$\hat{V}_k^q$	Phase q to neutral voltage magnitude estimate at bus k
P_h^q	Active power measurement at the network header through phase q
$\hat{P}_h^q$	Active power estimate at the network header through phase q
P_k^p	Phase p active power injection at bus k
$P_{(k,t)}^q$	Fraudulent power associated with phase q of the line connecting nodes k and t
ϵ	Voltage discrepancy threshold
γ	Trade-off weighting parameter

I. INTRODUCTION

Electricity fraud has always been one of the major concerns of electricity distributors. Its immediate negative effects are not only economic, but can also cause technical problems with enormous repercussions, such as interruptions in supply, a deterioration in the quality of the signal supplied to the customer causing damages in appliances, greater electricity losses in energy transmission, or worse still, endangering people's safety, among other things [1]. The deterioration of the sustainability of the entire electrical system as a result of electricity fraud is evident and a subject of debate in society [2].

There are different ways to steal electricity, the two most common being tapping electricity directly from the distribution network without passing through the meter or tampering with the energy meter [1]. The first of these two, which is the most widespread [3], consists of illegally connecting to a feeder at any point between the distribution transformer and the nodes where customers are connected (usually the leaf nodes). This type of theft is particularly difficult to detect if the illegal connection is not made immediately before the energy meter, but far away from it, in the middle of a power line away from customer connection points. Throughout this work, theft that involves a connection immediately before the meter is called *visible fraud*, while the connection far from any energy meter is named *invisible fraud*.

In the specialized literature, different criteria have been proposed to classify theft detection methodologies [3]-[12]. A first rough classification is based on whether or not new specific equipment to detect electricity theft is installed, i.e. hardware versus non-hardware solutions. Most of the proposed solutions are non-hardware, seeking to avoid incurring new investment costs. These non-hardware solutions are classified into three new categories: data-based, model-based, or hybrid methods. The main difference between data-driven and model-driven approaches is the use of network-related data by the second. Data-driven approaches have been very popular since the widespread use of smart meters (SMs), and concentrate

Submitted to the 24th Power Systems Computation Conference (PSCC 2026).

most of the non-hardware proposals from the specialized literature. However, these methodologies fail to detect *invisible fraud*, for which it is mandatory to use model-based or hybrid approaches. This work focuses on detecting and quantifying both visible and invisible fraud, so data-driven solutions are not considered the best techniques for addressing this problem.

Model-based solutions typically detect fraud by computing the deviation between estimated values and available measurements and comparing it to a preset threshold [3], [12]. Algorithms based on state estimators [13], [14], [15], [16], [17], [18], or power flow tools [19], [20], [21], [22], are used to estimate the quantities to be compared with the measurements. The greater the level of measurements collected, the better the performance of these model-based methodologies, with solutions based on state estimators performing better than those based on power flow [12]. However, the reality is far from this high degree of measurement redundancy, and although SMs have radically changed the observability of low-voltage networks [19], many more intermediate sensors should be installed for these model-based approaches to successfully detect and identify electricity fraud [16], mainly for power flow-based approaches [21], [22]. It is worth mentioning that to the best of the authors' knowledge, none of the model-based methodologies published to date addresses the detection and identification of invisible fraud and none of them take into account the specific nature of low-voltage lines, i.e., sections with three phases and four wires.

This paper proposes a methodology to address both visible and invisible fraud, leveraging advanced optimization techniques based on network model, sensor data at the distribution network, and SM readings. By constructing a linear model of the low-voltage (LV) network, the proposed approach identifies fraudulent connections. It aims to optimize the detection process, reducing both the time and cost involved in locating fraud incidents.

Thus, the main contributions of the work are:

- Propose a novel optimization-based methodology for detecting and locating visible and invisible electricity fraud in LV networks. The proposed solution specifically leads to a mixed-integer convex quadratic optimization problem, which guarantees that a global solution will be found.
- Leveraging network topology and connectivity data in conjunction with sensors and SM data to construct a more accurate linear model of the network. This linear model takes into account the unbalance nature of the three-phase four-wire low-voltage networks, non avoiding the consideration of the neutral or the coupling among phases.
- Reduce the time and operational costs associated with traditional fraud detection methods by optimizing field inspection efforts.

Section II motivates the addressed problem by analyzing the consequences of electricity fraud in a LV network. Section III formally state the problem. In Section IV the proposed

method is introduced. Section V evaluates the performance and robustness of the proposed solution under simulation and Section VI summarizes the main results of the conducted research and future work lines.

II. MOTIVATION OF THE PROBLEM

European LV distribution networks are inherently unbalanced systems, primarily composed of four-wire lines: the three phases (a , b , and c) and the neutral conductor (n). The networks supply a mix of single-phase loads, connected between one phase and the neutral, and three-phase loads, which are typically unbalanced. In recent years, the proliferation of private photovoltaic (PV) self-consumption installations at this voltage level has further altered traditional grid operation. PV sources operate alongside demand loads, which can cause reverse power flows at certain times across different parts of the grid, and occasionally even at the MV/LV transformer. Consequently, the resulting voltage profiles can differ significantly from well-known traditional patterns.

From an operational perspective, electricity fraud behaves as an unregistered, additional load. This practice is reflected in two main electrical effects. The first is an inconsistency in LV network energy balances, where loss percentages above a certain threshold may suggest the presence of non-technical losses in addition to technical ones. The second key indicator is an abnormal voltage profile along the grid, characterized by increasing voltage imbalances or unusually low-voltage values at certain nodes.

To illustrate the impact of fraud on voltages, the unbalanced 53-bus LV network 65037 from [23] is considered. A typical operating scenario is simulated with a three-phase illegal connection demanding 30 kW at bus 33. Although inherently three-phase, Fig. 1 depicts the phase-a voltage profile for simplicity. The green line represents the normal case (no fraud), while the red line shows the case with the illegal connection. A clear voltage drop appears at bus 33 and its electrically adjacent buses, potentially causing power quality issues, overloads in critical components such as power lines, and threats to supply continuity. This distinctive voltage deviation serves as a detectable signature for fraud identification and localization, highlighting the need for effective tools to uncover such unauthorized consumption.

III. PROBLEM DESCRIPTION

The work considered in this study uses a distribution network model defined by a graph $\mathcal{G}$, where the vertices represent the buses of the network, and the edges represent the electrical lines that interconnect them. Mathematically, the graph $\mathcal{G}$ is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of vertices or buses¹, and $\mathcal{E}$ is the set of edges or electrical lines that interconnect those buses. This work takes into account the unbalanced nature of LV distribution networks. Thus, the electrical quantities of the network at each point are

¹Note in a three-phase four-wire network a bus can be constituted by until 4 nodes, the three phases and the neutral

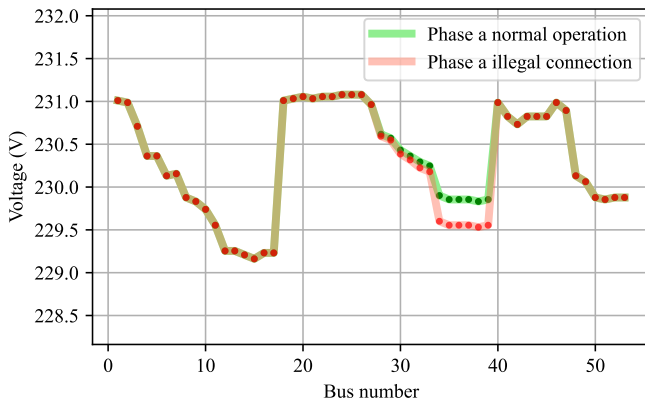


Fig. 1. Voltage profile (phase a) across the 53-bus LV network 65037 from [23] under normal conditions (green) and with a 30 kW illegal connection at bus 33 (red).

represented by four components corresponding to phases a , b and c and the neutral wire n .

The problem addressed in this work is the identification and location of illegal connections in LV distribution networks. An illegal connection is assumed to be attached to a power line between two adjacent buses k and t , where $(k, t) \in \mathcal{E}$. Only a subset of power lines is considered vulnerable to hosting illegal connections. Accordingly, the set $\mathcal{F} \subseteq \mathcal{V}$ is defined as the set of buses adjacent to those power lines that are candidates for hosting an illegal connection.

To approach the problem, the following hypotheses are considered:

- Information about the network topology and the types of conductors used is available. Therefore, network model $\mathcal{G}$ can be derived.
- Prior work has already been carried out to determine the phase assignment of each SM in the network.
- Synchronized measurements of active power, reactive power, and nodal voltages are available from each SM in the network, as well as from the distribution transformer at the head of the feeder.

It should be noted that these assumptions are increasingly realistic in modern distribution systems. Electricity distribution companies maintain detailed, geolocated records of all their electrical assets; combined with information on the conductor types used in the installation, this enables accurate knowledge of both the network topology and the electrical characteristics of its lines. Regulatory issues related to remuneration for the electricity distribution business [24] are one of the main reasons why distribution companies collect and keep all this information up to date. Likewise, synchronized “snapshots” of the distribution network are becoming increasingly common [25], [26], driven by utilities growing interest in detecting energy theft and other forms of non-technical losses. In particular, it is assumed that for each single-phase load connected to a bus k and measured by a SM between phase q and the neutral wire, time-aligned measurements of active power P_k^q , reactive power Q_k^q , and phase-to-neutral voltage V_k^q are simultaneously

collected. In addition, active and reactive power and nodal voltages are also collected at the low-voltage side of the distribution transformer, node h , by low-voltage supervisors [27], that is, P_h^q , Q_h^q and V_h^k .

IV. PROPOSED METHOD

The proposed method consists of two sequential stages as schemed in Fig. 2. In the first stage, candidate network segments potentially hosting fraud are preselected by comparing actual voltage measurements against predicted voltages derived from a reference network model, thus constructing $\mathcal{F}$; significant discrepancies between these values indicate areas warranting further investigation. The second stage refines this identification by precisely locating the fraudulent segments through the resolution of a novel optimization problem.

A. Identification of candidates areas to host fraud

As introduced in the problem description, the proposed approach relies on the electrical network model together with a set of measurements acquired from the system. Specifically, synchronized measurements of active power, reactive power, and nodal voltages are available from all SMs in the network, as well as from the feeder head. Based on this information, as it can be seen in Fig. 2, the proposed method proceeds in the following steps:

- 1) A model of the network is constructed based on the topology and the electrical parameters of the power lines.
- 2) The constructed model is fed with the active and reactive power measurements from the SMs, while the voltage at the head of the network is fixed to its measured value. Note that, at this stage, the nodal voltages measured by the SMs and the power measurements at the network head are not used.
- 3) Next, the voltages measured by the SMs are compared with those computed by the model-based power flow solution. Nodes for which this discrepancy exceeds a predefined threshold, i.e. $|V_k^q - \hat{V}_k^q| \geq \varepsilon^2$, are flagged as candidate buses where illegal connections may be present and, therefore are included in set $\mathcal{F}$.

Recall that the discrepancies in voltages for a simulated fraud in a network, discussed during the problem motivation in Section II, were already illustrated in Fig. 1.

B. Fraud detection and location

Once the set $\mathcal{F}$ has been constructed, the second stage of the algorithm begins. In this stage, the proposed method aims to identify both the location and magnitude of the illegal connections. To achieve this, two steps are performed. First, a linear model of the network is derived using sensitivity coefficients. Subsequently, an optimization problem based on that coefficients is formulated, whose solution yields the location and magnitude of the illegal connections in the LV network.

²Variables marked with a hat (e.g. $\hat{x}$) represent model estimates, while unmarked variables denote measured quantities.

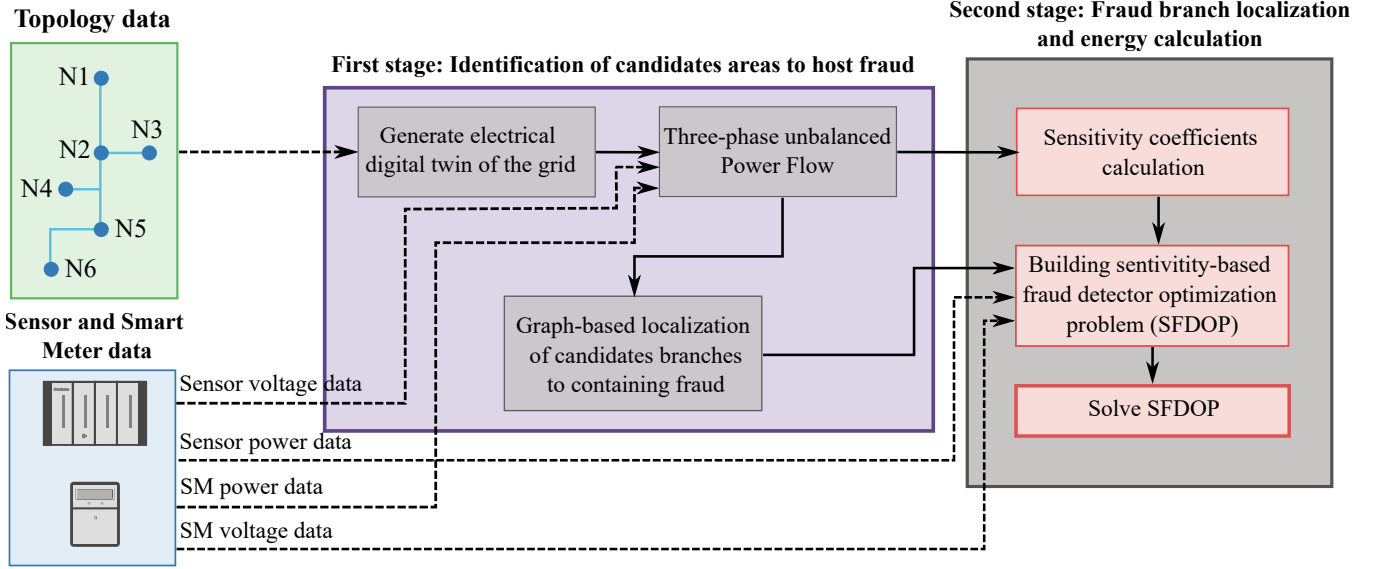


Fig. 2. Fraud detection and calculation methodology scheme.

The linear approximation of the network model is constructed by computing the sensitivity coefficients of nodal voltages with respect to power injection increments, i.e., $\partial \hat{V}_k^q / \partial P_t^p$, as well as the sensitivity of active power at the head of the network with respect to power injection increments, i.e., $\partial \hat{P}_h^q / \partial P_t^p$. The procedure to obtain these sensitivity coefficients is well described in [29], where their accuracy has also been evaluated. These sensitivities are obtained at an operating point of the network where the power flow was solved in the previous stage. The estimated values of nodal voltages, head power, and sensitivities at this operating point are denoted as $x|_0$ where x represent the specific magnitude. The use of sensitivity coefficients to linearize the network equations offers a significant advantage, as it allows formulating the constraints of the optimization problem to be solved by using linear equations. This avoids considering the exact network equations that would lead to a nonlinear and non-convex optimization problem. The proposed linear formulation makes the resulting methodology computationally efficient and robust.

Once the linear model of the network is constructed, the following optimization problem (1) is formulated:

$$\min \quad \gamma \sum_q \left(\hat{P}_h^q - P_h^q \right)^2 + \frac{1}{N_k} \sum_q \sum_{k \in \mathcal{V}} \left(V_k^q - \hat{V}_k^q \right)^2, \quad (1a)$$

$$\text{s.t.} \quad \hat{V}_k^q = \hat{V}_k^q|_0 + \sum_{t \in \mathcal{F}} \sum_p \frac{\partial \hat{V}_k^q}{\partial P_t^p} \Delta P_t^p, \quad (1b)$$

$$\hat{P}_h^q = \hat{P}_h^q|_0 + \sum_{k \in \mathcal{F}} \sum_p \frac{\partial \hat{P}_h^q}{\partial P_k^p} \Delta P_k^p, \quad (1c)$$

$$\Delta P_{(k,t)}^q = \Delta P_k^q + \Delta P_t^q, \quad (1d)$$

$$\delta_{(k,t)}^q \Delta P_{\max} \geq \Delta P_{(k,t)}^q \geq \delta_{(k,t)}^q \Delta P_{\min}, \quad (1e)$$

$$\sum_{(k,t) \in \mathcal{E}} \sum_q \Delta P_{(k,t)}^q \leq P_{\text{loss}}. \quad (1f)$$

In the above optimization problem, the decision variables are the power adjustments ΔP_k^q and the binary indicators $\delta_{(k,t)}^q$, defined for all nodes $k, t \in \mathcal{F}$ and phases $q \in \{a, b, c\}$.

The optimization problem aims to minimize the discrepancies between the voltage measurements at SMs and the corresponding model-based estimates, as well as between the estimated and measured active power at the head of the network (Both objectives are balanced by a trade-off weighting parameter γ). To this end, the variables ΔP_k^p are introduced, representing the additional power demand injections required at network nodes to supply undetected illegal connections. Note that these new power injections alters both the nodal voltages and the head power, as described by eqs. (1b) and (1c) respectively. The total fraudulent power associated with a line connecting nodes k and t is modeled as the sum of the power increments at its endpoints, i.e., $\Delta P_{(k,t)}^q$ (see eq. (1d)). Two additional constraints are included in the formulation. First, the fraudulent power on any line must exceed a predefined threshold; smaller values are discarded by setting the binary variable $\delta_{(k,t)}^q$ to zero (1e). Second, the total fraudulent power across the network is constrained to be less than the total losses of the system (including both technical and non-technical losses) as reflected in eq. (1f).

V. METHOD PERFORMANCE

To evaluate the performance of the proposed method, this section details the real-world network test case, the generated fraud scenarios, and the evaluation procedure.

A. Test case description

The performance of the proposed approach is assessed using a test case derived from adapted LV networks in northern Spain, originally detailed in [23]. These distribution systems consist of 22 independent radial LV networks, each fed by a 22/0.42 kV distribution transformer. While the foundational

network topologies, line parameters, and load profiles are documented in [23], the specific datasets employed in this study are available in a complementary, author-managed repository [28].

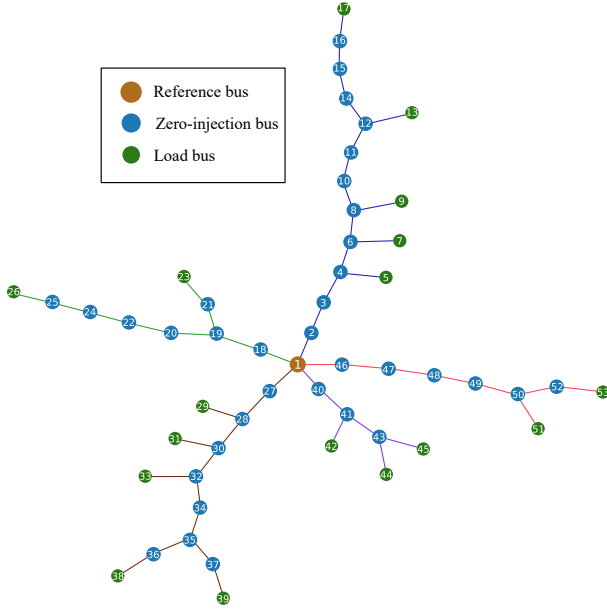


Fig. 3. LV network used in test cases.

From the available set, one network was selected and adapted to simulate an electricity fraud scenario. A single-line diagram of this network is illustrated in Fig. 3. In this diagram, green buses indicate consumer connection points (all equipped with SM measurement capabilities) while blue buses represent zero-injection buses without load or generation. The main characteristics of this network are summarized in Table I. The system comprises 53 buses, the majority of which include three phases and a neutral conductor, with a minority consisting of a single phase and neutral, resulting in a total of 208 nodes.

TABLE I
MAIN FIGURES OF THE TEST NETWORK

Feature	Value
Transformer rated power (kVA)	630
Extent (km)	0.87
Number of feeders	5
Number of loads	291
Number of buses	53
Number of nodes	208
Total Load (kVA)	261.47

B. Definition of fraud scenarios

To evaluate the effectiveness of the proposed methodology, four distinct fraud configurations are considered, with power demands ranging from 1 kW to 90 kW (representing approximately 0.38% to 34.4% of the total connected load). These scenarios are differentiated by two primary factors: the nature of the simulated fraud and the level of measurement visibility. Regarding the fraud type, each configuration is

modeled exclusively as either a single-phase or a three-phase power demand. The measurement visibility is determined by the status of the nodes defining the fraudulent line segment: high-visibility scenarios represent fraud occurring on a branch where, at least, one of the terminating nodes is equipped with SMs, while low-visibility scenarios represent fraud on a branch where the terminating nodes lack measurement support. Table II summarizes the specific configuration for each case.

TABLE II
FRAUD CASES CONFIGURATION

Case	Fraud Type	Measurement visibility	Fraud Segment
1	single-phase	high-visibility	32 - 33
2	single-phase	low-visibility	32 - 34
3	three-phase	high-visibility	32 - 33
4	three-phase	low-visibility	32 - 34

In scenarios characterized by high visibility (Cases 1 and 3), the fraud is simulated in a segment directly connected to buses equipped with SM measurements (bus 33). In contrast, scenarios characterized by low visibility (Cases 2 and 4) involve fraud in a segment not directly connected to SM-monitored buses (buses 32–34). In all instances, the fraud is modeled by introducing a power demand at a fictitious bus located within the affected branch.

C. Test results

This section presents the results obtained by implementing the proposed tool across the four cases defined previously. As introduced in Section IV, the methodology operates through two sequential stages:

- Stage 1: Potential candidate nodes and branches are identified by comparing field measurements, specifically voltage magnitude, against estimated values derived from the system model and power flow analysis. Any discrepancies exceeding the threshold ε are flagged as potential sites for fraud. In this section a value of $\varepsilon = 0.5$ has been considered.
- Stage 2: The optimization problem defined in (1) is solved to locate and quantify the existing fraud in the network. To ensure robustness, two constraints are applied to the optimization logic: a minimum power mismatch (ΔP_{min}) of 500 W is enforced, disregarding variations below this value as non-fraudulent noise, and the maximum fraud magnitude is set at the total measured network losses ($\Delta P_{max} = P_{loss}$), ensuring the solution remains physically consistent with the operational limits of the network.

The results for Case 1, representing single-phase fraud located near a SM, are summarized in Table III. The first column lists all candidate network branches identified during the Stage 1 of the methodology, while the remaining columns present the percentage of defrauded energy allocated to each branch during Stage 2, across a simulated fraud range of 1 kW to 30 kW. The branch corresponding to the simulated fraud location (32–33) is highlighted in bold to distinguish the ground truth. The results demonstrate the high effectiveness of

the proposed tool, as the correct segment consistently receives the highest percentage of allocated fraud energy. Performance improves with the magnitude of the fraud, with the method correctly identifying over 90% of the defrauded energy for power levels exceeding 5 kW. Although minor fractions of energy are occasionally allocated to adjacent sections (where the voltage drop patterns align with the objective function) the methodology successfully isolates the fraudulent segment with strong precision.

TABLE III
CASE 1: FRAUD ENERGY ALLOCATION (%) ACROSS CANDIDATE LV SECTIONS (SINGLE-PHASE, HIGH VISIBILITY).

$k, t \in \mathcal{F}$	1 kW	3 kW	5 kW	10 kW	20 kW	30 kW
27 - 28	0.0	19.1	0.0	0.0	6.7	6.8
28 - 29	0.0	0.0	0.0	0.0	0.0	0.0
28 - 30	0.0	0.0	0.0	0.0	0.0	0.0
30 - 31	0.0	0.0	0.0	0.0	0.0	0.0
30 - 32	0.0	0.0	0.0	0.0	0.0	0.0
32 - 33	53.7	75.7	73.0	90.2	90.3	91.9
32 - 34	0.0	0.0	0.0	0.0	0.0	0.0
34 - 35	0.0	0.0	0.0	0.0	0.0	0.0
35 - 36	0.0	0.0	0.0	0.0	0.0	0.0
36 - 38	22.2	0.0	0.0	0.0	0.0	0.0
35 - 37	0.0	0.0	0.0	0.0	0.0	0.0
37 - 39	0.0	0.0	0.0	0.0	0.0	0.0

Table IV presents the results for Case 2, which involves single-phase fraud occurring at a location with low measurement visibility, specifically, far from any SM connection points. Unlike Case 1, where proximity to an SM allows for high-precision identification, Case 2 exhibits a more distributed energy allocation. While the methodology successfully identifies the branch hosting the fraud (32–34, highlighted in bold) in most scenarios, it also allocates significant proportions of energy to adjacent segments, such as 32–33 and 35–36. This behavior highlights the inherent difficulty of detecting fraud in segments without direct SM observation, as the reduced voltage sensitivity at these locations makes distinguishing the fraudulent demand from surrounding network impacts more complex. Nevertheless, the correct branch generally retains the highest allocated energy percentage across the 1 kW to 30 kW fraud range, confirming the resilience of the tool even in low-visibility environments.

TABLE IV
CASE 2: FRAUD ENERGY ALLOCATION (%) ACROSS CANDIDATE LV SECTIONS (SINGLE-PHASE, LOW VISIBILITY).

$k, t \in \mathcal{F}$	1 kW	3 kW	5 kW	10 kW	20 kW	30 kW
27 - 28	0.0	10.7	0.0	0.0	0.0	0.0
28 - 29	0.0	0.0	0.0	0.0	0.0	0.0
28 - 30	0.0	0.0	7.5	0.0	7.2	7.1
30 - 31	0.0	0.0	0.0	0.0	0.0	0.0
30 - 32	0.0	0.0	0.0	0.0	0.0	0.0
32 - 33	0.0	0.0	39.2	23.7	40.7	24.0
32 - 34	0.0	37.8	50.8	67.8	51.7	68.6
34 - 35	0.0	0.0	0.0	0.0	0.0	0.0
35 - 36	81.3	47.4	0.0	0.0	0.0	0.0
36 - 38	0.0	0.0	0.0	0.0	0.0	0.0
35 - 37	0.0	0.0	0.0	0.0	0.0	0.0
37 - 39	0.0	0.0	0.0	0.0	0.0	0.0

Table V presents the results for Case 3, which models three-phase fraud under high-visibility conditions (close to an SM). Consistent with the findings in Case 1, the proximity to a measurement point significantly enhances the precision of the methodology, enabling accurate attribution of fraudulent power to the correct network segment (32–33, highlighted in bold). Across the entire simulated fraud range the proposed tool consistently allocates approximately 90% or more of the defrauded energy to the actual branch where the fraud occurs. Instances of incorrect allocation to adjacent sections are minimal, occurring in only one scenario with a negligible percentage. These results demonstrate that the algorithm effectively leverages the increased data availability of three-phase measurements to maintain robust detection accuracy even as the fraud magnitude scales.

TABLE V
CASE 3: FRAUD ENERGY ALLOCATION (%) ACROSS CANDIDATE LV SECTIONS (THREE-PHASE, HIGH VISIBILITY).

$k, t \in \mathcal{F}$	5 kW	10 kW	20 kW	30 kW	60 kW	90 kW
27 - 28	0.0	0.0	0.0	0.0	4.5	0.0
28 - 29	0.0	0.0	0.0	0.0	0.0	0.0
28 - 30	0.0	0.0	0.0	0.0	0.0	0.0
30 - 31	0.0	0.0	0.0	0.0	0.0	0.0
30 - 32	0.0	0.0	0.0	0.0	0.0	0.0
32 - 33	92.4	97.4	88.8	97.9	93.0	92.7
32 - 34	0.0	0.0	0.0	0.0	0.0	0.0
34 - 35	0.0	0.0	0.0	0.0	0.0	0.0
35 - 36	0.0	0.0	0.0	0.0	0.0	0.0
36 - 38	0.0	0.0	0.0	0.0	0.0	0.0
35 - 37	0.0	0.0	0.0	0.0	0.0	0.0
37 - 39	0.0	0.0	0.0	0.0	0.0	0.0

Table VI displays the results for Case 4, representing three-phase fraud in a low-visibility environment, that is, far from any measurement points. This scenario represents the most significant challenge for the proposed methodology, as evidenced by the dispersed energy allocation at lower fraud levels. The tool demonstrates robust performance only as the fraud magnitude increases, successfully isolating the correct network segment (32–34, highlighted in bold) when the fraud power exceeds approximately 11.5% of the total load. At lower power levels, the absence of direct SM observation complicates the distinction between the fraudulent demand and normal system variations, leading the algorithm to assign significant energy percentages to adjacent network sections. Nevertheless, the results confirm that the tool remains effective even in challenging, low-observability topologies provided that the anomaly is of sufficient magnitude.

To facilitate interpretation, Figs. 4 and 5 compare the allocation performance across all configurations. High-visibility scenarios (Cases 1 and 3) demonstrate superior precision, as the methodology concentrates allocation on the target segment with minimal error. Specifically, in High-visibility, Single-phase (Case 1) scenarios, the target segment receives 75.7–91.9% of the fraud energy, while High-visibility, Three-phase (Case 3) scenarios achieve 88.8–97.9%. Conversely, in Low-visibility scenarios (Cases 2 and 4), exact-branch precision decreases, yet neighborhood-level localization remains

TABLE VI
CASE 4: FRAUD ENERGY ALLOCATION (%) ACROSS CANDIDATE LV
SECTIONS (THREE-PHASE, LOW VISIBILITY).

$k, t \in \mathcal{F}$	5 kW	10 kW	20 kW	30 kW	60 kW	90 kW
27 - 28	0.0	4.3	0.0	7.4	4.7	1.5
28 - 29	17.1	0.0	0.0	0.0	0.0	0.0
28 - 30	0.0	0.0	1.3	0.0	0.0	0.0
30 - 31	6.8	0.0	7.9	0.0	2.8	5.4
30 - 32	0.0	13.1	0.0	0.0	0.0	0.0
32 - 33	23.5	31.3	42.5	40.7	29.7	14.4
32 - 34	0.0	0.0	0.0	51.8	62.6	78.4
34 - 35	0.0	0.0	16.4	0.0	0.0	0.0
35 - 36	19.1	16.6	31.9	0.0	0.0	0.0
36 - 38	33.4	34.1	0.0	0.0	0.0	0.0
35 - 37	0.0	0.0	0.0	0.0	0.0	0.0
37 - 39	0.0	0.0	0.0	0.0	0.0	0.0

robust. For Low-visibility, Single-phase (Case 2) scenarios, the true branch captures 37.8–68.6%, with combined allocations to the true and adjacent segments consistently exceeding 90% for fraud levels of 5–30 kW. Similarly, in Low-visibility, Three-phase (Case 4) scenarios, while reliable exact identification is limited to higher power levels (greater than 11.47% of total load), the combined allocation to the true and adjacent segments remains above 92%, effectively restricting the suspected area to a minimal number of LV sections.

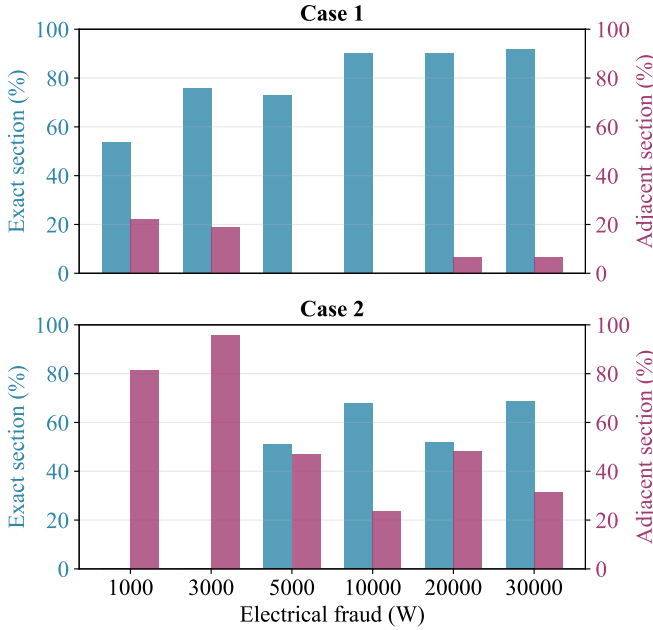


Fig. 4. Single-phase fraud detection results in Case 1 and Case 2 according to the fraud range.

D. Performance sensitivity to ε threshold

As introduced in Section IV, Stage 1 of the proposed methodology identifies potential fraud candidates by filtering network nodes based on a voltage discrepancy threshold, ε . This parameter serves as a critical input to the overall process: it defines the search space for the Stage 2 optimization problem, effectively filtering the candidate set and dictating

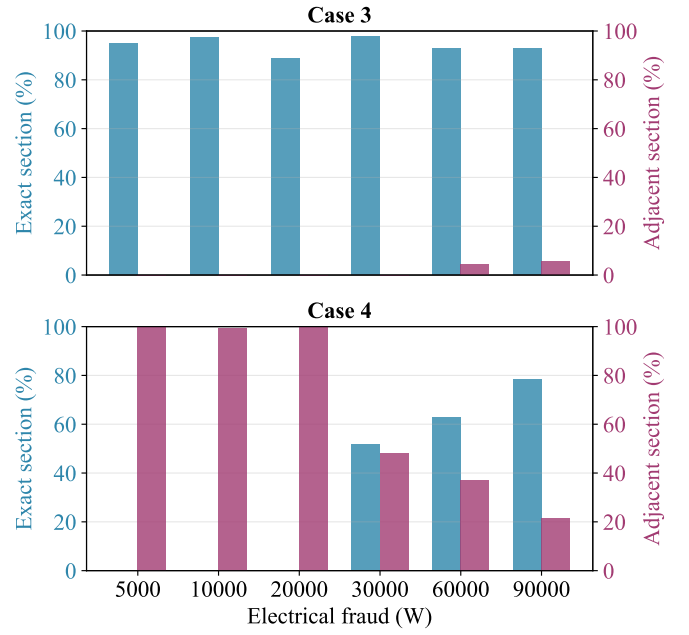


Fig. 5. Three-phase fraud detection results in Case 3 and Case 4 according to the fraud range.

both the computational complexity and the potential solution space. Consequently, the final energy allocation results are inherently sensitive to this threshold selection. To evaluate this impact, a sensitivity analysis was performed on Case 2 (20 kW, Single-phase, Low-visibility). Figure 6 illustrates the sensitivity of the fraud energy allocation to variations in ε , plotting the percentage assigned to the correct branch (32–34), the adjacent branch (32–33), and their combined attribution.

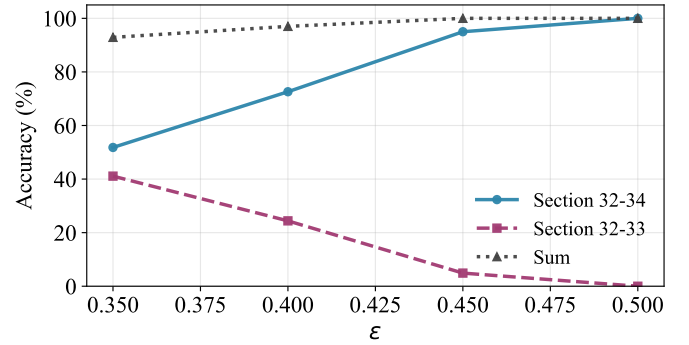


Fig. 6. Accuracy in locating sections where fraud is connected according to ε threshold value.

As shown in Figure 6, the threshold $\varepsilon = 0.35$ V provides strong neighborhood localization, successfully capturing the fraud within the immediate network vicinity, although exact branch identification remains spatially distributed. As ε increases, the Stage 1 filter becomes more restrictive, narrowing the search space for the Stage 2 optimization problem. This selectivity enhances spatial precision, achieving 100% exact branch identification at $\varepsilon = 0.5$ V. However, this sensitivity analysis highlights a critical design trade-off: while a higher

ε improves localization accuracy by minimizing candidate “noise”, an overly restrictive threshold risks excluding the true fraudulent branch entirely. Therefore, the selection of ε must balance candidate selectivity with the risk of omitting legitimate network sections.

VI. CONCLUSIONS

This paper introduces a novel methodology for the detection and localization of both visible and invisible electricity fraud by leveraging advanced optimization techniques based on detailed network models. Validated on real network data with synthetically generated illegal connection scenarios, the proposed approach demonstrates a high degree of effectiveness and consistently identifies the vicinity of fraudulent activity with remarkable precision. The two-stage structure provides an operationally meaningful workflow: an initial screening that reduces the search space to a limited set of suspicious LV sections, followed by an optimization stage that assigns fraudulent power to a small subset of candidate branches, enabling efficient prioritization of inspections. Although performance may degrade under limited observability, data inaccuracies, or when fraud is electrically distant from measurement points, the presented results and the ε sensitivity analysis confirm that the tool remains robust, typically concentrating the allocation between the true branch and adjacent ones. Overall, the method offers a practical and computationally efficient framework to support distribution operators in accelerating fraud localization while reducing fieldwork requirements. Future research will build upon this framework to address identified limitations, particularly the current susceptibility to measurement noise and topological inaccuracies. The primary advancement will involve transitioning from static, single-snapshot analysis to a multi-snapshot approach covering extended time windows to enhance statistical robustness and mitigate the impact of transient anomalies. Additionally, upcoming efforts will focus on adapting the methodology to accommodate modern grid topologies characterized by high levels of distributed generation penetration and integrating diverse data sources to further improve the diagnostic reliability of the tool and robustness against environmental and operational variations.

VII. ACKNOWLEDGEMENTS

This work was supported by the Ministerio de Ciencia e Innovación (MCIN) Agencia Estatal de Investigación (AEI) 10.13039/501100011033 through the ‘European Regional Development Fund (ERDF) A way of making Europe’ under Grant FlexOnGrid-PID2021-124571OB-I00.

REFERENCES

- [1] S. S. S. R. Depuru, L. Wang, and V. Devabhaktuni, *Electricity theft: Overview, issues, prevention and a SM based approach to control theft*, Energy Policy, vol. 39, no. 2, pp. 1007–1015, Feb. 2011.
- [2] Dianne Plummer, *Electricity Theft: Economic Burden And Sustainability Risks*, Forbes, Feb 21, 2025, 12:39am EST, <https://www.forbes.com/sites/dianneplummer/2025/02/21/electricity-theft-economic-burden-and-sustainability-risks/>
- [3] Z. Yan and H. Wen, *Performance Analysis of Electricity Theft Detection for the Smart Grid: An Overview*, in IEEE Transactions on Instrumentation and Measurement, vol. 71, pp. 1–28, 2022.
- [4] R. Jiang, R. Lu, Y. Wang, J. Luo, C. Shen, and X. Shen, *Energy-theft detection issues for advanced metering infrastructure in smart grid*, Tsinghua Sci. Technol., vol. 19, no. 2, pp. 105–120, Apr. 2014.
- [5] P. Glauner, J. A. Meira, and P. Valtchev, *The challenge of non-technical loss detection using artificial intelligence: A survey*, Int. J. Comput. Intell. Syst., vol. 10, no. 1, pp. 760–775, 2017.
- [6] J. L. Viegas, P. R. Esteves, R. Melício, V. M. F. Mendes, and S. M. Vieira, *Solutions for detection of non-technical losses in the electricity grid: A review*, Renew. Sustain. Energy Rev., vol. 80, pp. 1256–1268, Dec. 2017.
- [7] T. Ahmad, H. Chen, J. Wang, and Y. Guo, *Review of various modeling techniques for the detection of electricity theft in smart grid environment*, Renew. Sustain. Energy Rev., vol. 82, pp. 2916–2933, Feb. 2018.
- [8] G. M. Messinis and N. D. Hatziaargyriou, *Review of non-technical loss detection methods*, Electr. Power Syst. Res., vol. 158, pp. 250–266, May 2018.
- [9] A. Maamar and K. Benahmed, *Machine learning techniques for energy theft detection in AMI*, Int. Conf. Softw. Eng. Inf. Manage. (ICSIM), 2018.
- [10] M. S. Saeed et al., *Detection of non-technical losses in power utilities—A comprehensive systematic review*, Energies, vol. 13, no. 18, p. 4727, Sep. 2020.
- [11] Fernando de Souza Savian, Julio Cezar Mairesse Siluk, Tais Bisognin Garlet, Felipe Moraes do Nascimento, José Renes Pinheiro, Zita Vale, *Non-technical losses: A systematic contemporary article review*, Renewable and Sustainable Energy Reviews, Volume 147, 2021, 111205.
- [12] Guarda, F.G.K.; Hammerschmitt, B.K.; Capeletti, M.B.; Neto, N.K.; dos Santos, L.L.C.; Prade, L.R.; Abaide, A. *Non-Hardware-Based Non-Technical Losses Detection Methods: A Review*. Energies 2023, 16, 2054.
- [13] S. A. Salinas and P. Li, *Privacy-preserving energy theft detection in microgrids: A state estimation approach*, IEEE Trans. Power Syst., vol. 31, no. 2, pp. 883–894, Mar. 2016.
- [14] S.-C. Huang, Y.-L. Lo, and C.-N. Lu, *Non-technical loss detection using state estimation and analysis of variance*, IEEE Trans. Power Syst., vol. 28, no. 3, pp. 2959–2966, Aug. 2013.
- [15] S. Weckx, C. Gonzalez, J. Tant, T. D. Rybel, and J. Driesen, *Parameter identification of unknown radial grids for theft detection*, presented at the 3rd IEEE PES Innov. Smart Grid Technol. Eur. (ISGT Europe), 2012.
- [16] C. Carquex and C. Rosenberg, *Multi-timescale electricity theft detection and localization in distribution systems based on state estimation and PMU measurements*, presented at the 9th Int. Conf. Future Energy Syst., 2018.
- [17] H. L. Wong, C. K. Tan, W. -N. Tan, M. -T. Gan, S. -C. Yip and A. H. A. Bakar, *Energy Theft Identification: A State Estimation Model for Partial Meter Bypass*, in IEEE Access, vol. 13, pp. 121412–121431, 2025.
- [18] Rodrigo F.G. Sau, Tailan R. Garcia, Fernanda C.L. Trindade, Madson C. de Almeida, Bala Venkatesh, *Identification and ranking of customers with energy theft using hypothesis tests*, Electric Power Systems Research, Volume 225, 2023, 109862.
- [19] L. M. R. Raggi, F. C. L. Trindade, V. C. Cunha, and W. Freitas, “Non-technical loss identification by using data analytics and customer smart meters,” IEEE Trans. Power Del., vol. 35, no. 6, pp. 2700–2710, Dec. 2020.
- [20] M. Tariq and H. V. Poor, *Electricity theft detection and localization in grid-tied microgrids*, IEEE Trans. Smart Grid, vol. 9, no. 3, pp. 1920–1929, May 2018.
- [21] J. Y. Kim, Y. M. Hwang, Y. G. Sun, I. Sim, D. I. Kim, and X. Wang, *Detection for non-technical loss by smart energy theft with intermediate monitor meter in smart grid*, IEEE Access, vol. 7, pp. 129043–129053, 2019.
- [22] E. A. C. Aranha Neto and J. Coelho, *Probabilistic methodology for technical and non-technical losses estimation in distribution system*, Electr. Power Syst. Res., vol. 97, pp. 93–99, Apr. 2013.
- [23] X. Dominguez, A. Prado, P. Arbolea and V. Terzija, “Evolution of knowledge mining from data in power systems: The Big Data Analytics breakthrough”, Electric Power Systems Research, vol. 218, 109193, 2023.
- [24] Mateo, C., Pretico, G., Gómez, T., Cossent, R., Gangale, F., Frías, P., Fulli, G., 2018. *European representative electricity distribution networks*, Int. J. Electr. Power Energy Syst., 99, pp. 273–280, 2018.
- [25] A. L. Shah, A. T. Al-Awami and W. Mesbah, *Time Skew Mitigation in Smart Meters for Tampering Detection and Correction*, in IEEE Transactions on Instrumentation and Measurement, vol. 70, pp. 1–15, 2021.
- [26] F. B. B. Rolim, V. C. Cunha and F. C. L. Trindade, *Non-Technical Losses Detection and Location for LV Systems Applying Smart Meters*

- Measurements*, 2022 IEEE Power & Energy Society General Meeting (PESGM), Denver, CO, USA, 2022
- [27] López, I.G.; Zubia, H.U.; González, J.R.; Lartategui, I.A.; Sanz, M.H.; Sánchez, A.P. *low-voltage Supervision Systems: Technology, Applications, Use Cases And Deployment*. In Proceedings of the 25th International Conference on Electricity Distribution, Madrid, Spain, 3–6 June 2019; pp. 3–6.
- [28] <https://github.com/rubencarmona/lvnetworkdataset>
- [29] Rubén Carmona-Pardo, Rafael Morán-Corbacho, Álvaro Rodríguez del Nozal, Esther Romero-Ramos, *Practical sensitivity-based optimization technique to solve the hosting capacity problem in unbalanced low-voltage networks*. Sustainable Energy, Grids and Networks. Volume 45. 2026. 102140. ISSN 2352-4677. <https://doi.org/10.1016/j.segan.2026.102140>.